MSG No. 88.86 I_c4_1/a [Type IV, tetragonal]

Table 1: Wyckoff site: `8a`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[1,4,14,15]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{7}{8}\right]$	`[2,3,13,16]`
3	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[5,8,10,11]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	`[6,7,9,12]`
5	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[17,20,30,31]`
6	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{8}\right]$	`[18,19,29,32]`
7	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[21,24,26,27]`
8	$\left[\frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	`[22,23,25,28]`

Table 2: Wyckoff site: `8b`, site symmetry: `-4'..`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{7}{8}\right]$	`[1,4,30,31]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\right]$	`[2,3,29,32]`
3	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[5,8,26,27]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{7}{8}\right]$	`[6,7,25,28]`
5	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[9,12,22,23]`
6	$\left[0, \frac{3}{4}, \frac{5}{8}\right]$	`[10,11,21,24]`
7	$\left[\frac{1}{2}, \frac{1}{4}, \frac{5}{8}\right]$	`[13,16,18,19]`
8	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[14,15,17,20]`

Table 3: Wyckoff site: `16c`, site symmetry: `-1`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,14]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,15]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,10]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[7,11]`
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,13]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[12,16]`
9	$[0, 0, 0]$	`[17,21]`
10	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[18,30]`
11	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[19,31]`
12	$\left[0, \frac{1}{2}, 0\right]$	`[20,24]`
13	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[22,26]`
14	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[23,27]`
15	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[25,29]`
16	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[28,32]`

Table 4: Wyckoff site: `16d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[1,21]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[2,30]`
3	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[3,31]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,24]`
5	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[5,17]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[6,26]`
7	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[7,27]`
8	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[8,20]`
9	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[9,29]`
10	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[10,22]`
11	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[11,23]`
12	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[12,32]`
13	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,25]`
14	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[14,18]`
15	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[15,19]`
16	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[16,28]`

Table 5: Wyckoff site: `16e`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	`[1,28]`
2	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{4}\right]$	`[2,27]`
3	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{4}\right]$	`[3,26]`
4	$\left[\frac{3}{4},\ 0,\ z\right]$	`[4,25]`
5	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	`[5,32]`
6	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{3}{4}-z\right]$	`[6,31]`
7	$\left[\frac{1}{4},\ 0,\ \frac{3}{4}-z\right]$	`[7,30]`
8	$\left[\frac{1}{4},\ 0,\ -z\right]$	`[8,29]`
9	$\left[\frac{3}{4},\ 0,\ z+\frac{1}{2}\right]$	`[9,20]`
10	$\left[\frac{3}{4},\ 0,\ z+\frac{3}{4}\right]$	`[10,19]`
11	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{3}{4}\right]$	`[11,18]`
12	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[12,17]`
13	$\left[\frac{1}{4},\ 0,\ \frac{1}{2}-z\right]$	`[13,24]`
14	$\left[\frac{1}{4},\ 0,\ \frac{1}{4}-z\right]$	`[14,23]`
15	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}-z\right]$	`[15,22]`
16	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[16,21]`

Table 6: Wyckoff site: `16f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{4}\right]$	`[2,3]`
3	$\left[0,\ \frac{3}{4},\ -z\right]$	`[5,8]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}-z\right]$	`[6,7]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[9,12]
6	$\left[0,\ \frac{3}{4},\ z+\frac{3}{4}\right]$	[10,11]
7	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[13,16]
8	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-z\right]$	[14,15]
9	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[17,20]
10	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{3}{4}\right]$	[18,19]
11	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[21,24]
12	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}-z\right]$	[22,23]
13	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	[25,28]
14	$\left[0,\ \frac{3}{4},\ z+\frac{1}{4}\right]$	[26,27]
15	$\left[\frac{1}{2},\ \frac{1}{4},\ -z\right]$	[29,32]
16	$\left[0,\ \frac{1}{4},\ \frac{3}{4}-z\right]$	[30,31]

Table 7: Wyckoff site: 32g, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{3}{4}-y,\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$	[2]
3	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{1}{4}\right]$	[3]
4	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[y+\frac{1}{4},\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	[6]
7	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{3}{4}-z\right]$	[7]
8	$\left[x,\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	[10]
11	$\left[y+\frac{3}{4},\ \frac{3}{4}-x,\ z+\frac{3}{4}\right]$	[11]
12	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$	[14]
15	$\left[\frac{1}{4}-y,\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$	[15]
16	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{3}{4}-y,\ x+\frac{1}{4},\ z+\frac{3}{4}\right]$	[18]
19	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$	[19]
20	$\left[-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[20]
21	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[y+\frac{1}{4},\ \frac{3}{4}-x,\ \frac{1}{4}-z\right]$	[22]
23	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{1}{4}-z\right]$	[23]
24	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[25]
26	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	[26]
27	$\left[y+\frac{3}{4},\ \frac{3}{4}-x,\ z+\frac{1}{4}\right]$	[27]
28	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[28]
29	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[29]

continued ...

Table 7

No.	position	mapping
30	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{3}{4}-z\right]$	[30]
31	$\left[\frac{1}{4}-y,\ x+\frac{1}{4},\ \frac{3}{4}-z\right]$	[31]
32	$\left[x+\frac{1}{2},\ y,\ -z\right]$	[32]